

DIRECTIONS: THIS EXAMINATION CONTAINS A TOTAL OF **80** MULTIPLE CHOICE QUESTIONS. SELECT CAREFULLY THE BEST ANSWER AND BLACKEN ONLY THE LETTER OF YOUR CHOICE ON THE SEPARATE ANSWER SHEET PROVIDED

TIME ALLOWED: 2:00 HOURS

1. Which one of the following units of measurement is more convenient to express the size of cellular organelles?
A. Meter B. Centimeter C. Kilometer D. Micrometer
2. Which of the following step of the scientific method comes before all the rest?
A. hypothesis B. conclusion C. reporting result D. all
3. What do we call a scientist who specializes in studying insect?
A. An ecologist B. an entomologist
C. a Pala entomologist D. a microbiologist
4. To which one of the following organic molecules do enzymes belong?
A. carbohydrates B. protein C. vitamin D. lipids
5. Which process produces mRNA during protein synthesis?
A. translation B. replication C. mutation D. transcription
6. What is the molecule in plant cells that first captures the radiant energy from sun light?
A. ATP B. DNA C. Chlorophyll D. carbon dioxide
7. Which of the following laboratory glassware is used for culturing bacteria?
A. petri dish B. test tube C. filter paper D. measuring cylinder
8. Of the following which one is the main source from which plants get the nutrients necessary for their growth and development?
A. light B. chlorophyll C. atmosphere D. soil
9. Which of the following types of cancer is known to mostly develop in AIDS patients?
A. cervical cancer B. stomach cancer C. kaposi's sarcoma D. breast cancer
10. Which of the following is correct constitution of the sex chromosome of a normal woman?
A. XY B. XX C. XO D. XXY
11. What is the best term that express the movement of substances in cells against their concentration gradient? A. active transport B. passive transport
C. osmosis D. water
12. In cell division what is the phase that comes following the metaphase called?
A. Extra Phase B. Prophase C. Anaphase D. Telophase
13. Who was the person that first observed living cells moving around when he examined drops of under the microscope?
A. Robert Hook B. Anton Van Leeuwenhoek C. Theodor Schawnn D. All
14. The word science comes from Latin word Scientia meaning what?
A. study B. experiment C. life D. knowledge
15. Which of the following substances can be used over and over again in an enzyme catalyzed reaction? A. Enzyme-substrate complex B. substrate
C. Enzyme D. reaction product

16. Which of the following microorganisms are prokaryotic?
A. bacteria B. protozoa C. yeast D. algae
17. What is the intermediate compound that is formed during enzyme catalyzed reaction?
A. reactant B. substrate C. product D. enzyme-substrate complex
18. From which of the following disease can one protected by sleeping under the cover of a mosquito net?
A. sleeping sickness B. AIDS C. Malaria D. Athlete's foot
19. Which of the following parts of the flower did Mendel remove from young flowers to prevent self-pollination?
A. Stamens B. Sepals C. Petals D. Ovaries
20. What do we call the place where organisms live in their ecosystem?
A. abiotic B. biotic C. habitat D. all
21. Which of the following bases is not found in RNA?
A. Adenine B. guanine C. cytosine D. thymine
22. Which one of the following is different from the other three?
A. starch B. glycogen C. cellulose D. maltose
23. What happen when a plant absorbs in more water?
A. it shrinks B. it become turgid C. it becomes wilted D. it swells and bursts
24. Why do microorganisms decompose dead organic matters?
A. to release mineral nutrients for the plants B. to clean up the ecosystems
C. to drive energy for their own use D. to keep the balance of atmospheric gases
25. Of the following organic molecules which one includes a pentose sugar?
A. Glucose B. Fructose C. Galactose D. Deoxyribose
26. Through which of the following processes do all living organisms contribute to carbon cycle?
A. photosynthesis B. cellular respiration C. decomposition D. composition
27. In ecological context which of the following is population?
A. All living organisms in a habitat B. all the producers in a habitat C. all consumers in a habitat D. all individuals of the same species in the habitat
28. Through which process can inorganic ion from the soil be absorbed in to the root hairs?
A. dehydration B. diffusion C. active transport D. osmosis
29. How are organic molecules distinguished from in organic molecules? Organic molecules always contain?
A. oxygen hydrogen B. carbon and oxygen
C. oxygen and nitrogen D. carbon and hydrogen
30. Which industrial enzyme technology is lactase practically applied to produce lactose-free milk for use by lactose intolerant individuals?
A. chocolate manufacturing industry B. human hormone manufacturing industry
C. dairy product manufacturing industry
D. cereal food manufacturing industry.

31. Which kind of enzyme inhibitors bind to enzyme only weakly and the bond that hold them breaks easily releasing the inhibitor to allow the enzyme to become active again?
A. irreversible inhibitors B. reversible inhibitors C. inhibitors that bind strongly to enzyme D. inhibitors that permanently alter enzyme structure.
32. Of following activities which one is considered to have an indirect negative effect on bio diversity?
A. human and livestock population growth B. charcoal making and illegal logging C. deforestation for cultivation and urbanization D. monoculture farming with improved crop varieties
33. The instrument used to overcome the effect of gravity in plant growth experiment is called
A. thermostat B. potometer C. klinostat D. hydrometr
34. Hybrid vigour may be expressed by
A. production of normally viable seeds B. production of less resistant C. production of stronger and bigger plan D. production of plant less adapted than either parent
35. In a pea, the gene for green pod is dominant to the gene for yellow pod if one obtains atotal of 580 F2 pea plants , expected number of the two types of plant are
A. 428 green and 152 yellow B. 435 green and 145 yellow C. 340 green and 340 yellow D. 15 green and 430 yellow
36. One of the following statements is in correct
A. Enzymes work best in acidic PH B. Most human enzymes work best at body temperature C. Enzymes are proteins that undergo chemical change D. Co-enzymes are protein that work with enzyme
37. Which mammal loses more heat to the surrounding per gram body weight per day
A. A large- sized mammal B. A small- sized mammal C. A mammal with a small surface area to volume ratio D. A mammal with reduced surface area
38. In human nutrition the food substance that gives the most energy per gram of nutrient is
A. carbohydrates B. vitamins C. proteins D. fats
39. Suppose a double strand DNA molecule consisting of 400 bases contain total of 50 adenine, the total number of guanine expected in this molecule is
A.50 B.100 C.150 D.200
40. In certain plant RR genotype produces red and rr genotype produces white flowers. When the two genotypes are crossed they produce plants with pink flowers. Which of the following is expected if two pink flower producing plants are crossed?
A. All pink flower B. All red flower C. 1/4red, 1/2pink and 1/4white D. 3/4 pink and 1/4 white flower.
41. Which of the following chromosomal errors creates a cell with an extra chromosome?
A. non-disjunction B. inversion C. deletion D. duplication

42. What is the type of the scientific research that is directly concerned with studies of the way of gaining new knowledge related to human needs?
A. basic science B. applied science C. current science D. space science
43. In which of the following groups of organisms are all the members unicellular?
A. algae B. virus C. bacteria D. fungi
44. What is the function of DNA polymerase?
A. it hydrolyzes the DNA molecule B. it forms DNA by joining nucleotides
C. it converts the DNA to RNA D. it join two RNA molecules of form DBA
45. Of the following four elements which one is the least abundant in living things?
A. Oxygen B. Hydrogen C. Carbon D. Sulfur
46. Which of the following pairs of the processes add CO₂ to the atmosphere?
A. combustion and respiration B. respiration and photosynthesis
C. photosynthesis and fossilization D. decomposition and combustion
47. Which bacterial genus fixes nitrogen in the nodules of leguminous plants?
A. agrobacterium B. Escherichia C. rhizobium D. azobacter
48. Which of the following is the function of thyroxine? A. conversion of glycogen to glucose B. control of the metabolic rate C. development of sexual characteristics D. conversion of glucose to glycogen
49. Which of the following is an adaptation by desert plants such as cacti to conserve water?
A. having narrow and spiny leaves B. production of seeds that can be transported by water C. having broad leaves D. having leaves which are not covered by cuticle
50. What do you call glands that produce secretions which are directly released to the blood stream? A. endocrine B. salivary C. mammary D. exocrine
51. Which of the following is scientific name?
A. Canis simensis B. Abyssinian wolf C. Mountain nyala D. Gelada baboon
52. Part of respiratory system where gaseous exchange takes place is
A. bronchiole B. bronchus C. alveolus D. trachea
53. Blood vessels that carry deoxygenated blood and return it to heart are
A. lymph vessels B. capillaries C. arteries D. veins
54. What do we call a breeding technique which helps to get a combination of good traits from two different parents? A. cross breeding B. inter breeding C. selective breeding D. group breeding
55. Which of the following defects is different from the rest?
A. long sightedness B. short sightedness . C. astigmatism D. deafness
56. All of the following are chemical approaches to control microorganisms EXCEPT?
A. antibiotics B. disinfectants C. antiseptics D. autoclaving
57. Which of the following plant is endemic to Ethiopia?
A. Zea mays B. Eragrostis teff C. pinus radiata D. pinus sylvestris

58. What is biodiversity? It is the
 - A. measure of wealth of species in a place
 - B. abolishing of a stable ecosystem by human activities
 - C. nonrenewable natural resource we have
 - D. interaction of the biotic and a biotic components of the ecosystem.
59. Which of the following is true about reflex action? It is a
 - A. Learned response to stimulus
 - B. control response to the stimulus
 - C. fast response to the stimulus
 - D. slow response to the stimulus
60. The cell theory states that cells
 - A. Need food to produce energy
 - B. are the basic unit of life
 - C. remove poisonous waste products
 - D. are sensitive to changes in their surroundings
61. Which model of enzyme action requires that the shape of the substrate molecule is complementary to that of the active site?
 - A. The induced- fit model
 - B. The Lock-and key model
 - C. The activation energy model
 - D. The enzyme function model
62. If two parents with genotypes Aa are cross-bred and that there is no co-dominance in the inheritance pattern, what proportion of the offspring would have genotypes exactly like that of their parents
 - A.50%
 - B.75%
 - C.100%
 - D.25%
63. When an organism containing a gene which does NOT belong to it and is derived from somewhere else, the organism is said to be
 - A. hyrid
 - B. trangenic
 - C.mutant
 - D. clonal
64. Which of the following Hominids has the largest brain?
 - A. Homo sapiens
 - B. Homo erectus
 - C. Autralopithecus
 - D. Ardpithecus ramidus
65. A behavior that an organism is born with is called.
 - A. Innate behaviour
 - B. imprinted behaviour
 - C. Learned behaviour
 - D. Conditioned behavior
66. The sheep Dolly is an example of which biotechnological manipulation of animals by human?
 - A. cloned animal
 - B. Hybrid animal
 - C. Transgenic animal
 - D. Genetically engineered animal
67. Which of the following is an in organic molecule?
 - A. CaCo3
 - B. CH4
 - C.C6H12O6
 - D. C18H36O2
68. To which group of organic compounds do the triglycerides and waxes belong?
 - A. vitamins
 - B. Lipids
 - C. Protein
 - D. carbohydrates
69. What does an ethnologist study?
 - A. Soil type
 - B. Insect diet
 - C. Animal behaviour
 - D. fossil fuel
70. Which of the properties of water allows mosquito larva to hung themselves dawn into the water from the water surface?
 - A. Its high surface tension
 - B. Its high specific heat capacity
 - C. Its high latent heat vaporization
 - D. Its low density in solid state
71. What does the fermentation of glucose by yeast normally yield?
 - A. Lactic acid,co2, and2ATP
 - B. Alcohol, co2, and 2ATP
 - C. Alcohol, co2, and36ATP
 - D. Co2, H2O, and 36ATP

72. What is the highest level of ecological complexity from the alternatives listed below?
A. Community B. ecosystem C. population D. species
73. In a population growing at zero growth rate
A. the birth rate went to zero but the death rate remained constant B. the birth rate and the death rate remained equal to each other C. population will never grow at zero rate D. All
74. Which one of the following groups have the highest efficient of obtaining energy from grasses? A. primary carnivorous B. omnivorous C. herbivorous D. A&C
75. In which of the following alternatives are the animals endemic to Ethiopia?
A. walia ibex, hyena, zebra B. giraffe , zebra, camel C. wattles, ibis, elephant
D. mountain nyala, walia ibex, semen fox
76. One of the following is not way of reduced biodiversity?
A. deforestation B. conservation C. the impact of agriculture D. All
77. Biodiversity measured by
A. species richness B. ecological variability C. genetic variability D. All
78. In which phase population increases rapidly due to abundant resources?
A. log B. decline C. lag D. stationary
79. Which one are the correct sequences?
A. family – genus – species – order B. species – genus – family - order C. genus – family – order – species D. species – family – genus - order
80. Which one of the following is the medicinal enzyme used to treat cystic fibrosis
A. Abacavir sulphate B. pulmozyme C. Biochymosin D. Xylanase